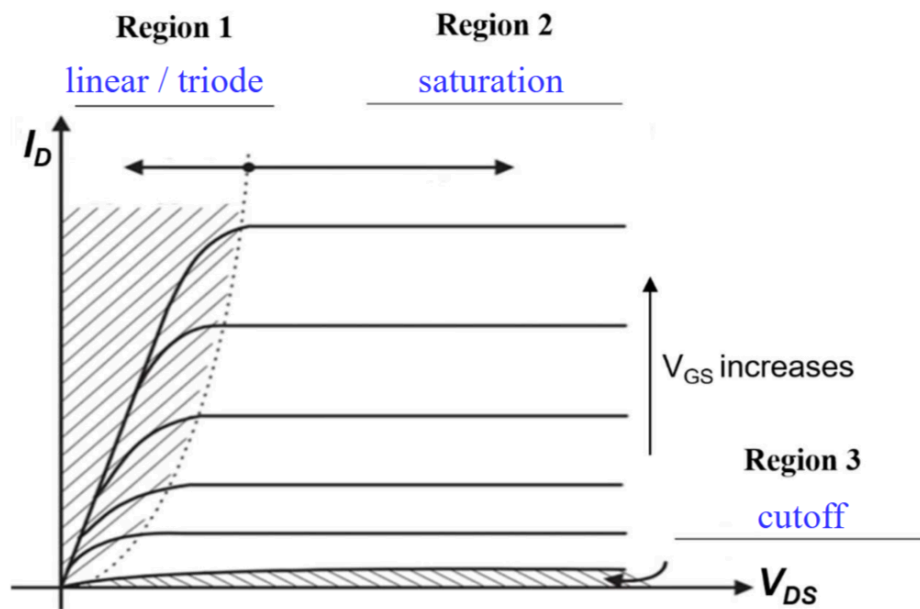


Q. 1 The figure below shows the I-V curve of a NMOS transistor,

(a) Name the 3 operation regions of the transistor in the figure.

(1 point)



Q1. The figure below shows the I-V curve of a NMOS transistor,

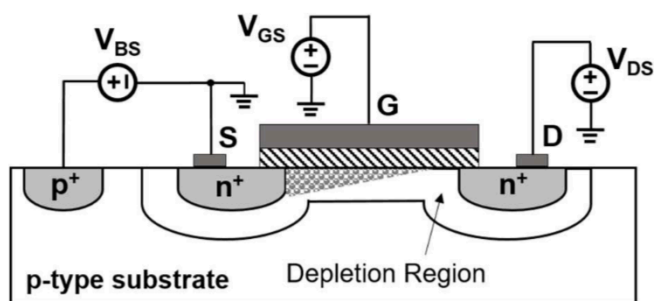
(b) To use this transistor as an amplifier, it should work in saturation region. To ensure the transistor works in this region, the bias should meet the following conditions
 $V_{GS} \geq V_{THN}$, $V_{DS} \geq V_{GS} - V_{THN}$ (2 points)

Operation Region	V_{GS}	V_{DS}	Remark
Linear (triode)	$\geq V_{THN}$	$\leq V_{GS} - V_{THN}$	Linear resistor
Saturation	$\geq V_{THN}$	$\geq V_{GS} - V_{THN}$	Constant I_D
Cut-off	$< V_{THN}$	–	$I_D \approx 0$

Q.2 The figure below shows the cross-section of a MOSFET device,

(a) The MOSFET shown below is operating in what region? (1 point)

Answer Saturation region



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When $V_{DS} \geq V_{GS} - V_{TH}$, channel is pinched-off

$$I_{D0} = \frac{1}{2} k'_n \frac{W}{L} (V_{GS} - V_{TH})^2 \rightarrow \text{independent of } V_{DS}$$

Q2. The figure below shows the cross-section of a MOSFET device,

(b) When the bulk-source voltage V_{BS} increases from zero to 300mV, the effective threshold voltage ~~increases~~ / **decreases** / ~~remains as constant~~ (delete as applicable) (1 point)

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The NMOS threshold voltage is given by:

$$V_{THN} = V_{TH0} + \gamma \left(\sqrt{|2\phi_F - V_{BS}|} - \sqrt{|2\phi_F|} \right)$$

V_{TH} **decreases** if SB junction is **forward biased**: $V_{BS} > 0$ (NMOS)

Q3. For a MOSFET working in saturation region, if $\lambda=0.01$, $r_o=5k\Omega$, $I_D=\underline{\hspace{2cm}20\text{ mA}\hspace{2cm}}$

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- When channel modulation effect is considered, I_D is also a function of V_{DS} .

$$I_D = \frac{1}{2} k'_n \frac{W}{L} (V_{GS} - V_{TH})^2 (1 + \lambda V_{DS})$$

- This means that there is an output resistance between drain and source, which can be expressed as

$$r_o = \frac{\partial V_{DS}}{\partial I_D} = \frac{1}{\lambda I_D} \quad \left(\lambda = \frac{\partial I_D / \partial V_{DS}}{I_D} \propto \frac{1}{L} \right) \quad \text{High } r_o \text{ can be obtained with long channel length } L$$
$$\rightarrow I_D = \frac{\partial V_{DS}}{\partial I_D} = \frac{1}{\lambda r_o}$$

Q4. For an NMOS device, when V_{DS} is changed from 0.6V to 0.9V, I_D varies from 3mA to 3.2mA. What will be the value of its r_o ?

Answer 1500 Ω

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- When channel modulation effect is considered, I_D is also a function of V_{DS} .

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$$r_o = \frac{\partial V_{DS}}{\partial I_D} = \frac{1}{\lambda I_D} \quad \left(\lambda = \frac{\partial I_D / \partial V_{DS}}{I_D} \propto \frac{1}{L} \right) \quad \text{High } r_o \text{ can be obtained with long channel length } L$$
$$\rightarrow r_o = \frac{\partial V_{DS}}{\partial I_D} = \frac{\Delta V_{DS}}{\Delta I_D} = \frac{0.3V}{0.2mA} = 1500\Omega$$

Q5. For a MOSFET device, V_{GS} is 0.7V, V_{TH} is 0.5V, $W/L=10$, $\mu_n C_{ox}=10\text{mA/V}^2$, its on resistance R_{on} is 50 Ω

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Drain Current Equation:
$$I_D = k'_n \frac{W}{L} \left[(V_{GS} - V_{TH})V_{DS} - \frac{1}{2}V_{DS}^2 \right]$$

where $k'_n = \mu_n C_{ox}$, W and L are channel width and length, respectively.

For small V_{DS} ($V_{DS} \ll V_{GS} - V_{TH}$), we can omit the second order term.

$$I_D \approx k'_n \frac{W}{L} (V_{GS} - V_{TH})V_{DS}$$

A linear dependence between I_D and V_{DS} , equivalent to a **linear variable resistance controlled by V_{GS}** . It is also a function of device geometry (W/L).

Define the channel resistance of transistor in linear region:

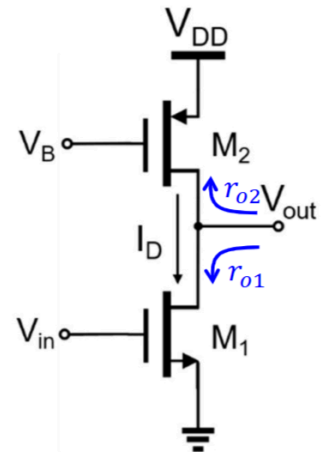
$$R_{DS} = \frac{V_{DS}}{I_D} = \frac{1}{k'_n \left(\frac{W}{L} \right) (V_{GS} - V_{TH})}$$

Q6. For a common source amplifier with current source load shown in the figure below, $I_D=10\mu\text{A}$, $V_{GS}-V_{TH} = 0.2\text{V}$, and $r_{o1} = r_{o2} = 100\text{ k}\Omega$, calculate amplifier's voltage gain. (2 point)

$$A_V = \frac{V_{out}}{V_{in}} = -g_{m1}(r_{o1} \parallel r_{o2})$$

$$g_m = \frac{\partial \left(\frac{1}{2} k'_n \frac{W}{L} (V_{GS} - V_{TH})^2 \right)}{\partial V_{GS}} = k'_n \frac{W}{L} (V_{GS} - V_{TH}) = \sqrt{2k'_n \frac{W}{L} I_D} = \frac{2I_D}{V_{GS} - V_{TH}}$$

$$A_V = \frac{V_{out}}{V_{in}} = -g_{m1}(r_{o1} \parallel r_{o2}) = -100\mu\text{S} \cdot (100\text{k} \parallel 100\text{k}) = -0.1\text{mS} \cdot 50\text{k} = -5$$

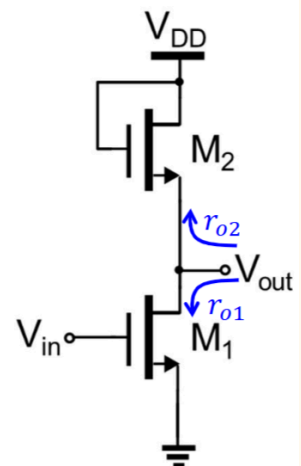


Q7. For a common source amplifier shown below,

(a) The load transistor works in saturation region. (1 point)

- M2 is **always in saturation** because the drain and the gate have the same potential.

$$V_{DS} > V_{GS} - V_{THN}$$

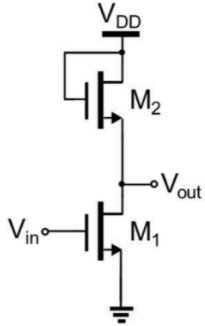


Q7. For a common source amplifier shown below,

(b) If the amplifier's bias current I_D increases, the amplifier's gain increases / decreases / remains the same (delete as applicable) (assume all transistors work in normal condition) (1 point)

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NMOS Diode Connected Load



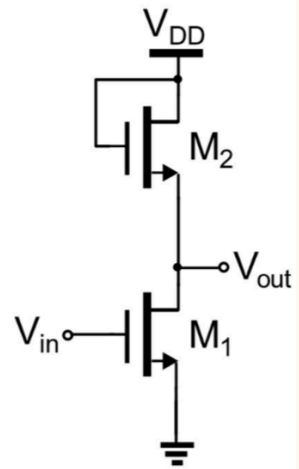
$$A_V = -g_{m1} \frac{1}{g_{m2} + g_{mb2}} = -\frac{g_{m1}}{g_{m2}} \frac{1}{1 + \eta}$$

$$A_V = -\frac{\sqrt{2k'_n \left(\frac{W}{L}\right)_1 I_{D1}}}{\sqrt{2k'_n \left(\frac{W}{L}\right)_2 I_{D2}}} \frac{1}{1 + \eta}$$

M1, M2 are in saturation region

$$A_V = -\frac{\sqrt{(W/L)_1}}{\sqrt{(W/L)_2}} \frac{1}{1 + \eta}$$

- The gain is independent of the bias current and voltage (so long as M1 remains in saturation region). → linear amplification



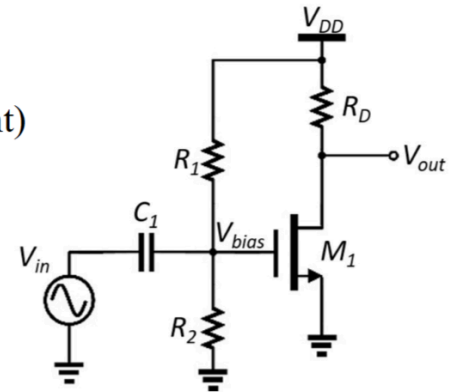
Q8. In the following circuit, transistor M1 operates in the saturation region. Assuming $\lambda=0$, $V_{TH}=0.5V$, $\mu_n C_{ox}=1mA/V^2$, $W/L=20$, $R_D=10k\Omega$, $V_{DD}=1.8V$.

(a) Find bias voltage V_{bias} that set the dc value of V_{out} to 0.9V. (1 point)

Answer 0.595V

$$I_D = \frac{V}{R} = \frac{0.9}{10k} = 90\mu A$$

$$I_D = \frac{1}{2} k'_n \frac{W}{L} (V_{bias} - V_{TH})^2 \rightarrow V_{bias} - V_{TH} = \sqrt{\frac{2I_D}{k'_n \frac{W}{L}}}$$

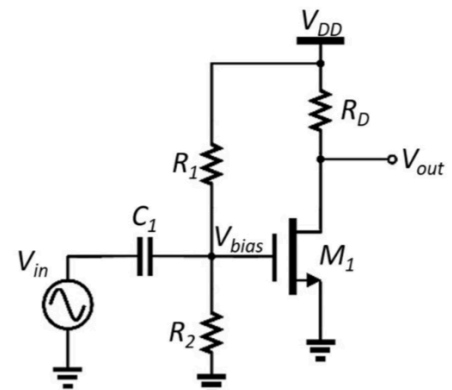


Q8. In the following circuit, transistor M1 operates in the saturation region. Assuming $\lambda=0$, $V_{TH}=0.5V$, $\mu_n C_{ox}=1mA/V^2$, $W/L=20$, $R_D=10k\Omega$, $V_{DD}=1.8V$.

(b) Find the amplifier small-signal gain $A_V=V_{out}/V_{in}$, (1 point)

Answer: -18.97

$$A_V = -g_m r_o = -k'_n \frac{W}{L} (V_{GS} - V_{TH}) (R_D || r_o)$$



Q9. The circuit below is a common source amplifier with a resistor load. The parasitic capacitors are considered in the amplifier bandwidth calculation.

(a) Identify the number of poles and zeros in this circuit (2 points)

Number of poles: 2

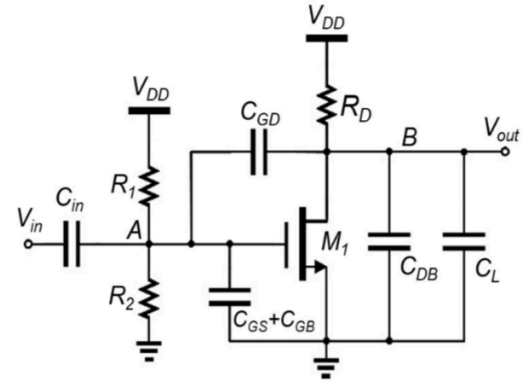
$$\text{Values of poles } \omega_1 = \frac{1}{(R_1 || R_2)[C_{GD}(1 + g_{m1}R_D) + C_{GS} + C_{GB} + C_{in}]}$$

$$\omega_2 = \frac{1}{R_D \left[C_{GD} \left(1 + \frac{1}{g_{m1}R_D} \right) + C_{DB} + C_L \right]}$$

Number of zeros: 2

Values of zeros: $\omega_1=0$, $\omega_2=g_{m1}/C_{GD}$

Apply Association of Poles with Nodes

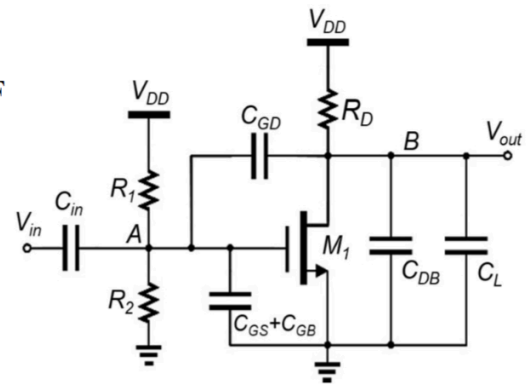


Q9. The circuit below is a common source amplifier with a resistor load. The parasitic capacitors are considered in the amplifier bandwidth calculation.

(b) If the dominant pole is at the output node **B**, if C_L increases from 2pF to 5pF, amplifier's bandwidth increases / decreases / remains constant (delete as applicable). (1 point)

$$\omega_2 = \frac{1}{R_D \left[C_{GD} \left(1 + \frac{1}{g_{m1}R_D} \right) + C_{DB} + C_L \right]}$$

$C_L \uparrow \quad \omega_2 \downarrow$



Q10. A current mirror circuit is shown below,

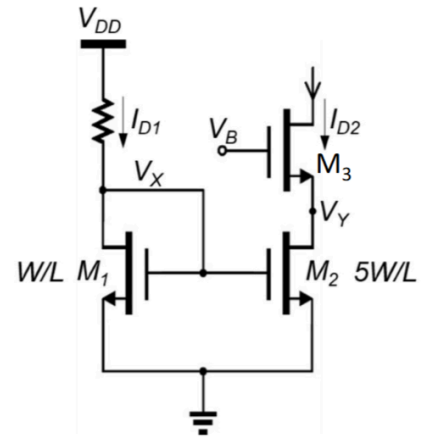
- (a) Give the condition that can completely remove systematic gain error in the circuit. (1 point)

Answer 1: $V_Y = V_X$

Answer 2: $V_X = V_{GS1}$

$$V_Y = V_B - V_{GS} = V_X = V_{GS}$$

$$V_B = V_{GS} + V_{GS}$$



Q10. A current mirror circuit is shown below,

- (b) Explain the reason for your answer to question (a). (2 points)

Systematic gain error definition: $E_{rr} = \frac{I_{OUT} - I_{OUT,ideal}}{I_{OUT,ideal}}$

Assume $\lambda_1 = \lambda_2$

$$\frac{I_{out}}{I_{in}} = \frac{I_{D2}}{I_{D1}} = \frac{\left(\frac{W}{L}\right)_2 (1 + \lambda V_Y)}{\left(\frac{W}{L}\right)_1 (1 + \lambda V_X)} \quad I_{out} = \frac{\left(\frac{W}{L}\right)_2 (1 + \lambda V_Y)}{\left(\frac{W}{L}\right)_1 (1 + \lambda V_X)} I_{in}$$

Gain error:

$$I_{out,ideal} = \frac{(W/L)_2}{(W/L)_1} I_{in}$$

$$E_{rr} = \frac{I_{out} - I_{out,ideal}}{I_{out,ideal}} = \frac{(1 + \lambda V_Y)}{(1 + \lambda V_X)} - 1 = \frac{\lambda}{1 + \lambda V_X} (V_Y - V_X)$$

If $\lambda V_X \ll 1$ $E_{rr} \approx \lambda (V_Y - V_X)$

Systematic gain error can be completely eliminated by making $V_Y = V_X$

